



A
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A Hybrid Multi-Stage Machine Learning System for
Accessible Thyroid Pre-Screening
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DECLARATION

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Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

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ABSTRACT

Thyroid disorders are a common yet often overlooked health issue. Identifying these disorders is complicated, especially in rural areas with limited resources. Specialists and lab tests are often unavailable, making timely diagnosis difficult. Early and accurate identification is crucial, as it leads to better patient outcomes and reduces the strain on the healthcare system. This article suggests a two-stage, multimodal machine learning framework for the early screening of thyroid disorders.

The framework combines three methods: visual examination of neck images, symptom-related questionnaires, and optional thyroid hormone profiles (TSH, T3, T4). In Stage 1, a lightweight convolutional neural network (CNN) analyzes clinical neck images to predict the likelihood of goiter, hypothyroidism, hyperthyroidism, and thyroid nodules. A symptom-scoring module enhances the prediction's accuracy. Certain cases are flagged immediately, while more uncertain predictions advance to Stage 2. In this second stage, the hormone test values are integrated with symptom data to clarify ambiguous hormone patterns using a calibrated decision model. Initial tests on a pilot dataset show that this multimodal approach significantly improves prediction stability compared to using either images or hormones alone. The hybrid method increases both overall accuracy and system resilience, addressing some of the issues with laboratory testing. The system is designed as a user-friendly, point-of-care screening tool that is accessible for all cases and not meant to replace diagnostic evaluations, focusing on early detection in underserved regions. This framework marks a significant step toward community-level thyroid screening on a large scale. It also provides opportunities for connection with mobile health platforms, telemedicine, and public health screening programs.

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LIST OF ABBREVIATIONS

- **AI:** Artificial Intelligence
- **CNN:** Convolutional Neural Network
- **DL:** Deep Learning
- **ML:** Machine Learning
- **RF:** Random Forest
- **SVM:** Support Vector Machine
- **T3:** Triiodothyronine
- **T4:** Thyroxine
- **TSH:** Thyroid Stimulating Hormone
- **UGC:** University Grants Commission

CHAPTER 1

INTRODUCTION

1.1 Introduction

The thyroid gland is one of the key endocrine glands in the human body. It regulates metabolic processes by producing thyroid hormones. Located in the front of the neck just below the larynx, the thyroid generates hormones like triiodothyronine (T3) and thyroxine (T4). These hormones are essential for controlling metabolism, growth, energy production, and organ function. They affect nearly every physiological process, including heart rate, body temperature, digestion, and energy use.

Thyroid disorders occur when the thyroid gland produces either too much or too little hormone. These conditions can disrupt normal body function and lead to various health issues. The most common thyroid disorders include hypothyroidism, hyperthyroidism, thyroid nodules, and goiter. Hypothyroidism arises when the thyroid doesn't produce enough hormones, causing symptoms like fatigue, weight gain, cold intolerance, and a slow metabolism. Hyperthyroidism, on the other hand, occurs when excessive thyroid hormones are produced, which can result in weight loss, rapid heartbeat, anxiety, and heat intolerance.

Thyroid nodules are another common issue. These abnormal growths within the thyroid can be benign, but some may require medical evaluation due to potential health risks. A goiter is an abnormal enlargement of the thyroid gland, often apparent as swelling in the neck. While goiters often do not cause significant symptoms, they may indicate underlying thyroid problems that need assessment.

Thyroid disorders pose a significant global health challenge. Epidemiological studies estimate that millions worldwide suffer from thyroid-related diseases. In India, over forty million people are thought to be affected. The prevalence is particularly high among women, the elderly, and those in iodine-deficient regions. Due to the variety of symptoms and the complexity of hormone regulation, thyroid disorders frequently go undiagnosed for long periods. This can lead to delayed treatment and potential complications.

Early detection and prompt diagnosis of thyroid disorders are essential for effective management. However, traditional diagnostic methods mainly rely on laboratory hormone tests, imaging techniques like ultrasound, and consultations with specialists.

These methods may not be easily accessible, especially in rural or low-resource settings. As a result, many people may remain unaware of their thyroid issues until they worsen. .

Recent advancements in artificial intelligence (AI), machine learning (ML), and deep learning (DL) offer new opportunities to enhance healthcare diagnostics. Machine learning can identify complex patterns in large datasets, making it suitable for medical predictions. Deep learning models, such as convolutional neural networks (CNNs), have excelled in medical image analysis, allowing for automated disease detection through visual data.

Combining machine learning with clinical features and physiological data presents a promising approach to developing intelligent healthcare screening systems. Such systems can assist healthcare professionals in making preliminary assessments and risk predictions, enabling earlier diagnoses and better patient outcomes.

This project aims to create a hybrid machine learning system for early thyroid disorder screening. The system incorporates various information sources, including neck images, symptom-based questionnaires, and hormone test values, to provide a thorough risk assessment. By integrating deep learning techniques with traditional machine learning models, the proposed system seeks to deliver an accessible and cost-effective solution for preliminary thyroid screening, especially in areas with limited medical resources.

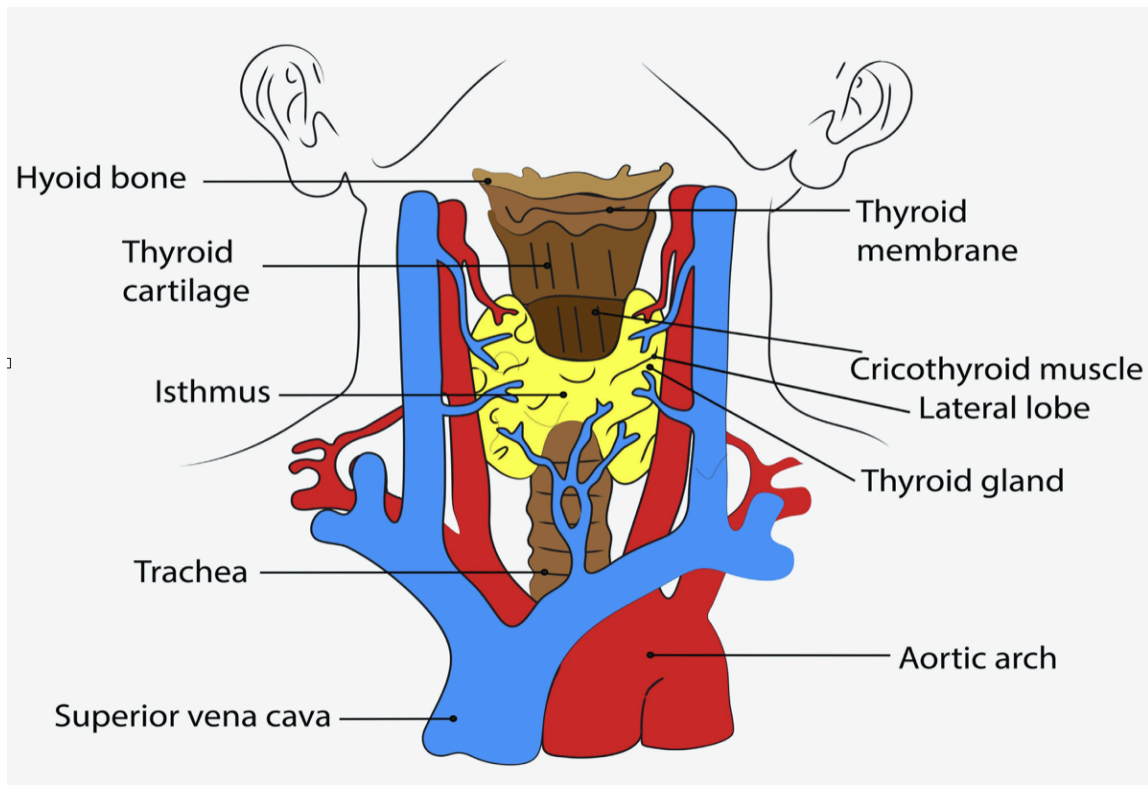


Fig. 1.1 - Anterior View

1.2 Project Description

The primary goal of this project is to design and implement an intelligent thyroid pre-screening system that combines both machine learning and deep learning techniques. This system aims for early risk assessment of thyroid disorders by analyzing various information sources, such as visual cues, patient symptoms, and biochemical data.

Traditional thyroid diagnosis primarily depends on lab tests measuring thyroid hormone levels, including Thyroid Stimulating Hormone (TSH), triiodothyronine (T3), and thyroxine (T4). While these tests provide reliable diagnostic data, they require laboratory facilities and trained medical staff. In many rural or low-resource healthcare settings, access can be limited. Consequently, individuals may delay seeking medical attention, worsening their symptoms and raising the risk of complications.

The proposed system addresses these challenges with a two-stage hybrid screening framework that combines image analysis and clinical symptom evaluation to estimate the likelihood of thyroid disorders. The first stage conducts primary screenings by analyzing external neck images and patient-reported symptoms. A convolutional neural network model based on the EfficientNetB0 architecture extracts significant visual features from neck images and classifies potential thyroid conditions.

Alongside image analysis, the system employs a symptom-based questionnaire to assess common thyroid-related symptoms, including fatigue, sensitivity to temperature, weight changes, neck swelling, and other signs of thyroid dysfunction. Each symptom is assigned a weight based on its clinical relevance, and responses go through a scoring system to produce a preliminary risk score.

The results of the image model and the symptom questionnaire are combined using a weighted fusion strategy to create an initial risk assessment. This first stage acts as the primary screening tool, allowing the system to identify individuals who may need further evaluation.

If the predicted risk falls into an intermediate or borderline range, the system transitions to the second analytical stage. Here, thyroid hormone test values are added. A Random Forest machine learning model processes the hormone data and provides a refined prediction of thyroid conditions. The integration of image analysis, symptom assessment, and hormone data develops a comprehensive multimodal screening framework that improves prediction reliability.

The proposed system serves as a pre-screening tool rather than a replacement for clinical diagnosis. Its goal is to provide an accessible and cost-effective way to identify individuals at risk for thyroid disorders and encourage them to seek professional evaluation. By using machine learning technology, the system aims to ease the burden on

healthcare facilities and enhance early detection rates, particularly in underserved regions.

Another important aspect of the project is its focus on accessibility and user-friendliness. The system can be accessed through a web-based interface or a mobile platform. Users can upload neck images, fill out symptom questionnaires, and optionally enter hormone test results. The system processes this information and generates a risk assessment along with suitable recommendations.

Creating such an intelligent screening system demonstrates how artificial intelligence can revolutionize healthcare diagnostics. By integrating multiple data sources and utilizing advanced machine learning techniques, the proposed solution provides a scalable and practical method for the early detection of thyroid disorders.

In summary, this project introduces a hybrid, multi-stage machine learning framework that combines deep learning-based image analysis, symptom-based risk assessment, and hormone-based classification to form an accessible thyroid pre-screening system. The system aims to support early detection, improve healthcare accessibility, and lay the groundwork for future research in AI-assisted medical diagnostics.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Diagnosing and predicting thyroid disorders is a significant area of study in endocrinology and medical informatics. Thyroid diseases are a major healthcare challenge due to their high prevalence and the complicated nature of accurate diagnosis. Traditionally, doctors use physical exams, lab hormone tests, and imaging techniques like ultrasound scans to diagnose these conditions. While these methods are reliable, they often require specialized equipment and trained medical staff, which may not always be available in rural or resource-limited settings.

In recent years, the rapid growth of artificial intelligence (AI), machine learning (ML), and deep learning (DL) has led to the development of diagnostic systems that help healthcare professionals. These technologies are increasingly used in medical imaging, disease prediction, and clinical decision support systems. Machine learning models can analyze complex datasets and spot patterns that traditional statistical methods might miss.

Many researchers have worked on applying AI models to detect thyroid diseases. Several studies have looked into using machine learning algorithms to predict thyroid disorders based on biochemical test results, clinical symptoms, and imaging data. Deep learning models, in particular, have shown strong results in medical image analysis, allowing for automated detection of abnormalities in imaging data.

This chapter reviews the current research on thyroid disease detection using machine learning and deep learning techniques. The focus is on various approaches, including hormone-based predictive models, ultrasound-based imaging systems, symptom-based diagnostic methods, and hybrid multimodal systems that integrate multiple data sources. The purpose of this literature review is to identify the strengths and weaknesses of these

existing methods and to highlight the research gap that informs the development of a proposed hybrid thyroid screening system.

2.2 Machine Learning in Medical Diagnosis

Machine learning has become a valuable tool in healthcare analytics. It allows for the creation of predictive models that assist in detecting and diagnosing diseases. These algorithms learn from past data to identify patterns that can predict outcomes from new data. They have been successfully applied to various medical areas, including cancer detection, predicting cardiovascular disease, diagnosing diabetic retinopathy, and detecting neurological disorders.

One major benefit of machine learning in medical diagnosis is its ability to handle large amounts of clinical data and extract valuable insights. Medical datasets often involve complex relationships between variables, such as physiological measures, lab test results, and patient symptoms. Machine learning models can evaluate these relationships and identify correlations that traditional analysis might overlook.

Supervised learning algorithms are often used in medical prediction systems. These algorithms are trained with labeled datasets, where each sample has a known outcome. After training, the model can classify new samples based on learned patterns. Common supervised learning algorithms in medical diagnosis include Support Vector Machines (SVM), Decision Trees, Random Forests, Logistic Regression, and Artificial Neural Networks.

Random Forest is one of the most commonly used machine learning algorithms in healthcare. It combines several decision trees to improve prediction accuracy and prevent overfitting. Random Forest models are particularly effective with tabular datasets that include clinical and biochemical data since they can capture nonlinear relationships and handle noisy data well.

Another important category of machine learning models in healthcare is neural networks. Artificial Neural Networks are modeled after the human brain and consist of interconnected nodes that process information across multiple layers. Deep learning models, which have many hidden layers, have achieved impressive results in tasks like image recognition, speech processing, and medical imaging.

Machine learning has significantly improved diagnostic accuracy in healthcare and led to automated screening systems. However, many current machine learning systems rely on

structured data, like lab test results, which may not always be accessible in all healthcare settings. This highlights the need for alternative methods that incorporate various information sources, including images and symptom data.

2.3 Thyroid Disease Detection Using Hormone-Based Models

Thyroid function is usually assessed through lab tests that measure thyroid-related hormones in the blood. The most common hormone indicators are Thyroid Stimulating Hormone (TSH), triiodothyronine (T3), and thyroxine (T4). These hormone levels provide important insights into thyroid gland function and are frequently used in clinical diagnosis.

Several studies have explored using machine learning models to predict thyroid disorders based on hormone test data. One earlier method applied decision tree algorithms to classify patients based on hormone levels and clinical characteristics. Decision trees create interpretable models that help doctors understand the factors involved in diagnosis.

Support Vector Machines (SVM) have also been commonly used to classify thyroid diseases. SVM models work effectively with high-dimensional datasets and can identify optimal decision boundaries that separate different disease classes. Research has shown that SVM-based models can achieve high accuracy in predicting thyroid conditions using hormone datasets.

Logistic regression is another widely used model for predicting thyroid diseases. It estimates the probability of a disease based on input features and provides interpretable coefficients that indicate the impact of each variable. While logistic regression is straightforward and computationally efficient, its performance may be limited when dealing with complex nonlinear relationships.

Random Forest classifiers have become popular for predicting thyroid diseases due to their reliability and ability to manage intricate datasets. Random Forest models combine multiple decision trees and aggregate their predictions for more consistent results. Studies have shown that Random Forest models can reach accuracy levels above 80% when applied to thyroid hormone datasets.

Despite the effectiveness of hormone-based machine learning models, these methods rely heavily on lab test data. In many rural or underserved areas, patients may not have access to regular blood tests due to financial constraints or limited medical facilities. Therefore,

there is a need for alternative screening methods that can identify potential thyroid disorders before lab tests are conducted.

2.4 Deep Learning for Medical Image Analysis

Deep learning has changed the field of computer vision and is essential for analyzing medical images. Convolutional Neural Networks (CNNs) are specialized deep learning models tailored for image processing tasks. CNNs automatically learn hierarchical features from images, enabling them to detect patterns and structures within the visual data.

In medical imaging, CNNs have been effectively used for tasks such as tumor detection, organ segmentation, and disease classification. These models can analyze complex visual patterns and identify abnormalities that may not be easily seen by the human eye.

A major advantage of CNN-based models is their ability to automatically extract features. Traditional image analysis methods require manual feature engineering, where experts develop specific features to describe images. In contrast, CNNs learn relevant features directly from the data during training.

Transfer learning is a common technique in deep learning that allows models trained on large datasets to be adapted for specific tasks. Pretrained models like VGGNet, ResNet, Inception, and EfficientNet are often used in medical image analysis. These models are initially trained on large image datasets like ImageNet and can be fine-tuned with smaller medical datasets.

EfficientNet is a modern deep learning architecture that provides better performance while being computationally efficient. EfficientNet models use a compound scaling method that balances network depth, width, and resolution for optimal performance. EfficientNetB0, the smallest variant, is popular in transfer learning tasks due to its high accuracy and relatively low computational requirements.

In thyroid disease detection, most deep learning research has concentrated on analyzing ultrasound images of the thyroid gland. Ultrasound imaging provides detailed information about the thyroid's structure and texture, helping to identify nodules and other abnormalities. CNN-based models have achieved high accuracy in classifying thyroid nodules from ultrasound images.

However, ultrasound imaging requires specialized equipment and trained technicians, which may not be accessible in every healthcare setting. This limitation has prompted

researchers to explore alternative screening methods that use more easily obtainable images, such as external neck photographs.

2.5 Symptom-Based Clinical Scoring Systems

Diagnosing thyroid disorders depends on clinical symptoms, which play a vital role in diagnosis. Thyroid dysfunction can lead to multiple symptoms that affect a person's metabolic processes and hormonal system. Symptoms include fatigue, weight changes, temperature sensitivity, mood disturbances, and cardiovascular issues.

Several clinical scoring systems have been developed to help physicians diagnose thyroid disorders based on patient symptoms. The Billewicz Index is a well-known scoring system that medical professionals use to assess the risk of hypothyroidism. The Billewicz Index assigns numerical scores to various symptoms and clinical findings, enabling physicians to estimate the likelihood of thyroid dysfunction.

Diagnostic systems that use symptoms for diagnosis offer effective initial screenings since they do not require special equipment, and results can be obtained quickly. However, these systems have limitations, as they rely on patients to provide their medical information. Many symptoms of thyroid disorders are nonspecific and can occur in various other medical conditions.

Machine learning techniques have been used to improve symptom-based diagnostic models by assigning optimized weights to various symptoms and identifying symptom combinations that strongly relate to specific diseases. Creating better predictive models requires integrating symptom data with information from imaging studies and laboratory tests.

2.6 Multimodal Machine Learning Systems

Multimodal machine learning systems use different types of data to improve their ability to predict outcomes. In healthcare, these systems combine clinical data, medical images, lab test results, and patient history to create a thorough evaluation of a patient's health status.

Combining various data types can significantly improve diagnosis accuracy since each type offers useful but different information. Imaging data can show structural issues, while lab tests provide biochemical insights into how the body functions. Symptom data captures the patient's personal experience and can serve as early signs of disease. Several recent studies have explored using multimodal machine learning systems to predict diseases.

These systems often employ ensemble learning techniques or decision-level fusion methods to integrate predictions from different models. Multimodal systems generally deliver greater accuracy and reliability compared to single-modality models because they leverage the strengths of multiple models.

For example, multimodal systems may use ultrasound images, hormone test results, and clinical features together to assist in diagnosing thyroid disease. However, many current multimodal systems still rely on specialized medical imaging methods that might not be accessible in resource-poor areas.

2.7 Research Gap

Even though machine learning has advanced significantly in detecting thyroid disease, some challenges still need addressing. Much of the existing research focuses primarily on either hormone-based classification or ultrasound image analysis. While these methods perform well in controlled environments, they may not be practical for screening large populations.

Hormone-based models require lab testing, which can be expensive and time-consuming. Ultrasound-based models also need specialized imaging equipment and trained professionals, making them harder to use—especially in rural or under-resourced areas.

Additionally, many machine learning models currently rely on a single type of data, which may not capture the full spectrum of thyroid issues. Single-modality models might struggle to accurately classify borderline cases where symptoms, hormone levels, and imaging findings do not match.

To tackle these challenges, we need a screening system that is user-friendly and flexible. It should be able to draw information from various sources while reducing reliance on pricey medical infrastructure. A hybrid multimodal system that includes image analysis, symptom evaluation, and optional hormone data could effectively screen for thyroid issues early on.

The system proposed in this project aims to fill this research gap by creating a two-stage hybrid machine learning framework. This system employs deep learning to analyze pictures of the neck, uses risk scoring based on symptoms, and optionally includes hormone data to enhance prediction accuracy. The goal is to provide a cost-effective and scalable method for early thyroid disease screening, particularly in resource-limited healthcare settings.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Overview of the Proposed System

The proposed system presents a mixed approach. It addresses current challenges by developing a user-friendly screening framework that leverages machine learning to analyze easily accessible information.

The system features a two-step screening process. The first round involves analyzing external neck images and symptoms reported by the patient. The image analysis module employs a deep learning model based on the EfficientNetB0 architecture to detect signs of thyroid issues in the neck area, such as swelling or asymmetry. Meanwhile, a questionnaire module evaluates symptoms potentially related to thyroid dysfunction.

To establish an initial risk score, the outputs from the image model and the questionnaire are combined using a weighted fusion strategy. This score helps determine whether the prediction is confident enough or if further testing is necessary.

The second stage of analysis starts when the predicted risk score falls into a borderline range. Thyroid Stimulating Hormone (TSH), triiodothyronine (T3), and thyroxine (T4) are some of the hormone values considered in this stage. A Random Forest classifier processes these biochemical markers, enhancing predictions and offering a clearer overview of thyroid health. By tying together these different testing methods, the system creates a strong and adaptable screening process that can function effectively across diverse healthcare settings. This proposed method improves diagnostic reliability by combining visual evidence, clinical symptoms, and biochemical data into a single decision-making framework.

3.2 System Architecture

This design incorporates various modules that allow for a comprehensive assessment of a patient's risk by evaluating different areas of health. The modules work together to generate a unified data source from several locations, enabling an accurate risk evaluation for individual patients.

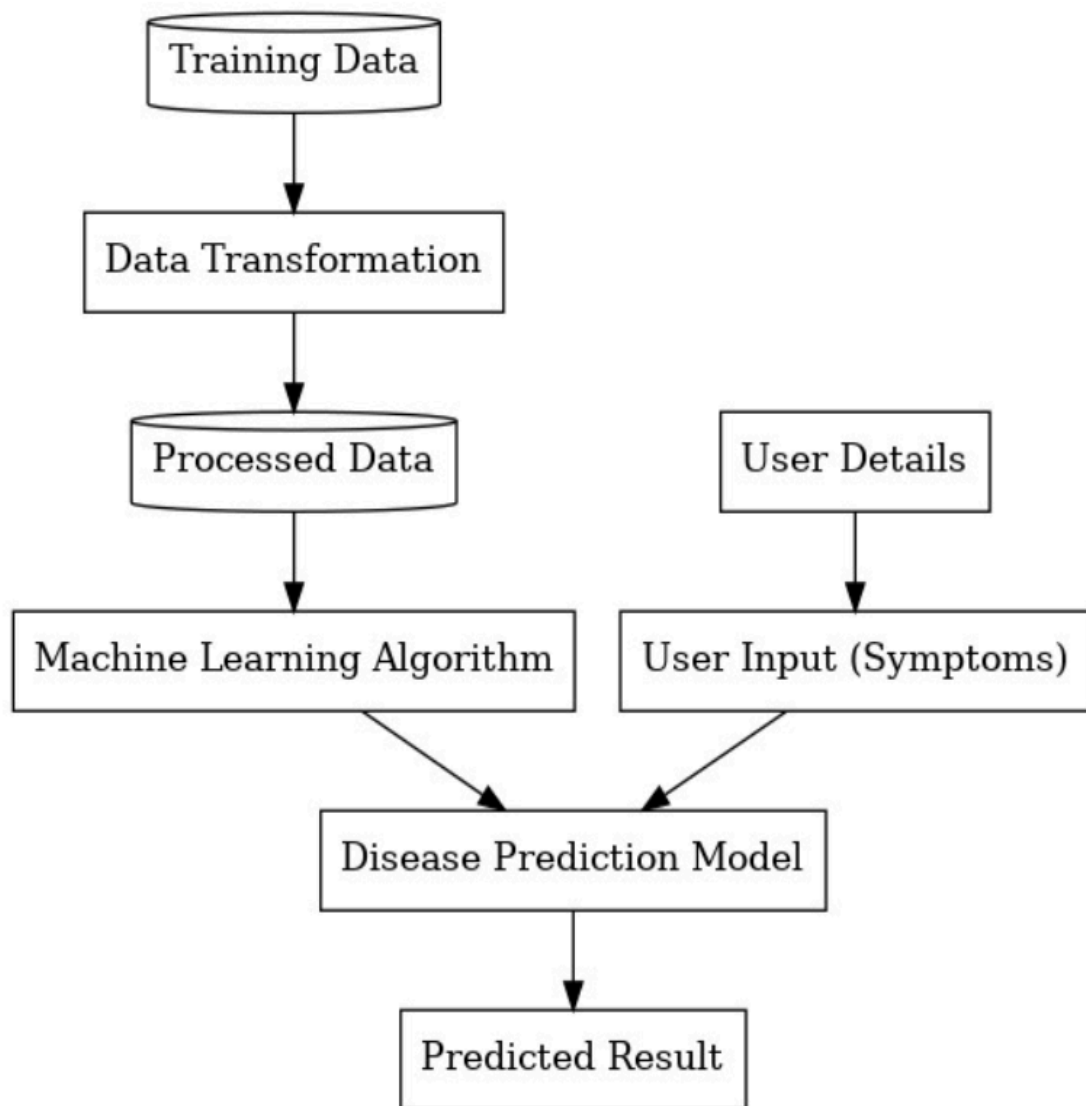


Fig. 3.1 System Architecture Diagram

The major components of the system are :

1. Data acquisition module
2. Image preprocessing module
3. Deep learning image classification model
4. Symptom questionnaire evaluation module
5. Stage-1 fusion mechanism
6. Hormone analysis module
7. Final prediction and recommendation module

The process begins when the user uploads a neck image and completes a symptoms questionnaire. The deep learning model analyzes the image for signs of thyroid problems, while the questionnaire assesses symptoms associated with thyroid disorders. Next, the system combines both sets of results to develop an initial risk score. If the system is confident in its prediction, it provides immediate recommendations. However, if the results are ambiguous and the risk level appears moderate, the system requests hormone test results and further examines the case with another machine learning model.

Ultimately, the system consolidates all analyses and test results to deliver a final prediction, risk level, and suggested next steps.

3.3 Data Collection

Data collection is essential for building any machine learning system in healthcare. If the data is not reliable—high quality, diverse, and accurately representative—the predictive models will not be effective. For the thyroid screening system, the goal is to gather various data types to cover all aspects of thyroid health. This includes neck images, responses from symptom questionnaires, and results from biochemical hormone tests.

This system employs a multimodal approach, integrating these different information types to reinforce predictions. Each data type contributes its unique perspective. Images

reveal the thyroid's physical appearance, symptom responses reflect the patient's daily experience, and hormone values provide biochemical context.

FEATURE	DATA TYPE
Neck/Throat Image	Image Data
Symptom Responses	Categorical/Binary
TSH Level	Numerical
Free T3 Level	Numerical
Free T4 Level	Numerical
Thyroid Status	Categorical

Table 3.1 Summary of Dataset Features

Obtaining sufficient high-quality data presents a challenge—public thyroid datasets are limited. Thus, the data collectors pull from various sources: public medical datasets, online image libraries, and even synthetic clinical data generated for training models. Combining these sources helps the system identify real patterns relevant for detecting thyroid problems.

3.3.1 Image Dataset Collection

The primary dataset for the image-based system comprises photos showing the neck, focusing on the area where the thyroid is located. The thyroid gland resides at the front lower part of the neck. When issues occur, such as goiter or thyroid nodules, you might notice swelling or other visible changes.

This dataset includes both normal and abnormal thyroid images, organized into categories based on thyroid conditions:

- Normal thyroid
- Swelling from hypothyroidism
- Structural changes from hyperthyroidism
- Thyroid nodules or goiter

Including images of healthy necks is crucial because the model needs to understand what “normal” looks like to avoid misclassifying everything as abnormal. Without these normal examples, the model risks signaling problems for all images.

Images come from diverse sources, including medical repositories, clinical databases, and curated collections about thyroid conditions. Since these images are taken in different settings, they vary significantly—some sharp, others blurry, with different lighting, angles, and backgrounds. While this variety might introduce noise, it actually helps the model learn to manage the range of images it could encounter in real-world scenarios. To keep everything organized, images of each class are stored in separate folders. This setup allows machine learning tools to easily find the correct label for each image during training.

3.3.2 Image Dataset Challenges

Collecting data for thyroid imaging poses challenges. The major issue is the lack of many labeled thyroid image datasets, especially compared to fields like radiology or dermatology. If you search for public datasets with labeled thyroid images, particularly simple neck photos, you will likely find limited options. Most datasets available focus on ultrasound images. While ultrasound reveals what is happening inside the thyroid, it requires expensive equipment and skilled operators. What this project needs is a simpler solution : images you can snap with a phone.

Then there’s the issue of image quality. When you pull photos from the internet, they vary widely, with different lighting, resolutions, and random elements in the background. If you don’t process them correctly, those differences can confuse machine learning models. That’s why, before feeding anything to the model, the images undergo cleanup and enhancement to fix inconsistencies and help the model learn to handle variety.

Despite these challenges, the dataset ends up with enough visual diversity for the model to recognize real patterns related to thyroid issues.

3.3.3 Symptom Questionnaire Data

The structured symptom questionnaire gathers medical data for each participant and will also help identify other thyroid disorders. Many thyroid disorders come with physiological symptoms due to hormonal imbalances in the body. Therefore, the clinical information collected through the questionnaire can provide valuable insights about a thyroid disorder prior to conducting laboratory tests.

The structured symptom questionnaire lists various symptoms that can occur with thyroid disease, with each symptom linked to a specific question. Participants indicate whether each symptom is present or absent on the questionnaire.

Examples of symptoms linked with hypothyroidism are :

- Low energy levels
- Increased sensitivity to cold temperatures
- Unexplained weight gain
- Dry or rough skin
- Constipation
- Depression or mood changes
- Slow heart rate
- Muscle weakness

Symptoms commonly associated with hyperthyroidism include:

- Anxiety or nervousness
- Heat intolerance
- Unexplained weight loss
- Tremors
- Rapid or irregular heartbeat
- Excessive sweating
- Difficulty sleeping

Symptoms associated with thyroid nodules or goiter may include:

- Visible swelling in the neck
- Difficulty swallowing
- Hoarseness in voice
- Neck pain
- Breathing difficulty

The answers are converted into simple yes or no codes to show whether each symptom is present. Each symptom is assigned a number based on its importance for thyroid problems. These special numbers are used to calculate a score that indicates the likelihood of having a problem. This score is part of the overall check-up process.

Using a list of questions about symptoms helps the system incorporate what doctors understand about thyroid problems while predicting what might be happening. This method is really useful when images alone do not provide enough information to make an accurate decision about someone's condition.

3.3.4 Hormone Dataset Collection

The second stage of the proposed system includes thyroid hormone test results if such data is available. These biochemical indicators offer important information about thyroid gland function.

The primary hormone markers considered in the system are :

- Thyroid Stimulating Hormone (TSH)
- Triiodothyronine (T3)
- Thyroxine (T4)

These hormone levels are commonly used in clinical practice to diagnose thyroid disorders. Abnormal hormone levels might indicate excessive or insufficient thyroid activity.

Typical reference ranges for these hormones are approximately:

- TSH: 0.4 – 4.0 mIU/L
- T3: 80 – 200 ng/dL
- T4: 4.5 – 12.0 µg/dL

When hormone values significantly deviate from standard ranges, they can suggest thyroid dysfunction. For example, elevated TSH levels alongside low T3 and T4 levels may indicate hypothyroidism, while low TSH levels coupled with high T3 and T4 levels could signal hyperthyroidism.

In the proposed system, hormone data is inputted into a Random Forest classifier that refines predictions made during the first stage of screening. This stage only occurs when the initial prediction is uncertain, ensuring that hormone testing happens only when necessary.

3.3.5 Multimodal Dataset Integration

Finally, the system combines all three types of data: images, symptom questionnaire responses, and hormone test results. Each type offers distinct advantages. Images show physical signs of issues, symptoms reflect the patient's feelings, and hormones provide biochemical evidence.

Bringing them together enhances the reliability of predictions. Models relying only on one type of information do not match the effectiveness of this integrated approach.

3.4 Data Preprocessing

Raw data from different sources can seem impressive, but it usually needs preparation before being useful for machine learning. Preprocessing addresses inconsistencies and improves quality, making everything suitable for model training.

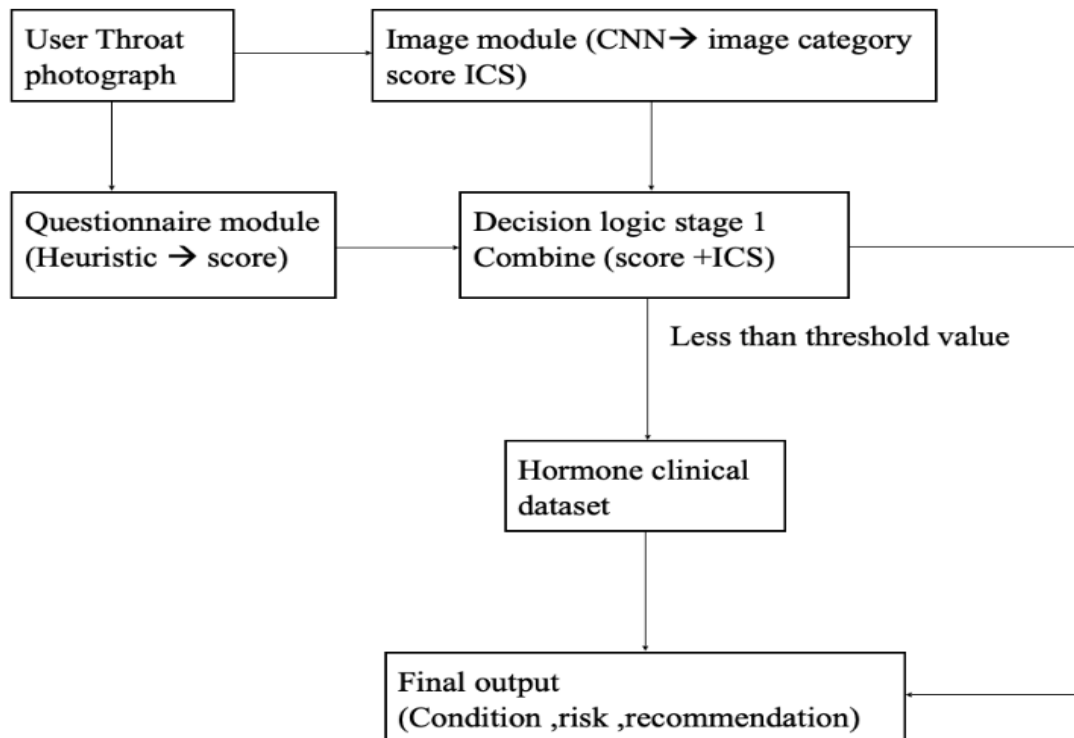


Fig. 3.2 Data Flow Diagram

3.4.1 Image Resizing

All images undergo a resizing process to 224×224 pixels. This is the input size that EfficientNetB0 requires, maintaining consistency for the model.

3.4.2 Image Normalization

Image Normalization Next, pixel values are normalized by dividing them by 255 to compress them into the range of 0 to 1. This step reduces variations in pixel intensity and speeds up training.

3.4.3 Data Augmentation

Data augmentation generates new, slightly altered versions of the original images to expand the training set. Simple techniques such as rotating, flipping, zooming, changing brightness, and adjusting width or height help the model learn to accommodate images taken under different conditions. Essentially, it makes the model less sensitive to variations in appearance.

3.4.4 Dataset Splitting

The entire dataset is divided into three parts: 70% for training, 15% for validation, and 15% for testing. Training helps the model learn, validation monitors for overfitting and tunes hyperparameters, and testing provides a final evaluation.

3.5 Deep Learning Model for Image Analysis

This system employs EfficientNetB0, a deep learning model built for high accuracy without requiring excessive computational resources. It belongs to the EfficientNet family, recognized for balancing model depth, width, and resolution rather than just increasing a single parameter.

3.5.1 Convolutional Neural Networks in Medical Imaging

Convolutional Neural Networks (CNNs) are commonly used in medical imaging. They excel at detecting patterns that a human eye might overlook, such as faint edges or minor

abnormalities. CNNs operate by stacking convolutions, pooling operations, and nonlinear activation functions.

The structure comprises convolutional layers (detecting raw features), activation functions (introducing nonlinearity), pooling layers (reducing dimensions), and fully connected layers (integrating features for predictions). As the network goes deeper, it captures more complex structures. Pooling reduces the amount of data the network has to process and helps control overfitting.

The final fully connected layers convert all extracted features into a diagnosis. In practice, CNNs have made significant contributions—identifying tumors, outlining organs, and flagging abnormalities. Their ability to recognize complex patterns is particularly valuable in thyroid imaging, where details are crucial.

3.5.2 EfficientNet Architecture

EfficientNet transformed how we scale neural networks. Instead of solely deepening or widening networks, it balances depth, width, and input resolution simultaneously. This balance yields better accuracy with less computation compared to older models.

These models range from B0 to B7. B0 serves as the baseline: it offers solid accuracy without demanding extensive resources. That's why B0 is used here—it performs very well without requiring high-end hardware.

3.5.3 Transfer Learning Strategy

Medical imaging faces a significant challenge—there aren't large labeled datasets like those available for animal or vehicle photos. Training a deep network from scratch is often impractical. Transfer learning provides a solution.

You start with a model pre-trained on a large general dataset (in this case, EfficientNetB0 pre-trained on ImageNet), freeze the initial convolutional layers so they remain unchanged, and add a classification head for thyroid images. The base layers recognize edges, shapes, and textures generally; then, the new layers learn to identify features specific to health.

Initially, the base layers are frozen. Only the new custom layers for thyroid data undergo training during this phase.

3.5.4 Model Architecture Design

The architecture consists of two main parts: EfficientNetB0 as the backbone and a custom classification head.

EfficientNetB0 extracts high-level features from the input image. The classification head follows, structured as:

- ❖ A global average pooling layer that condenses each feature channel into a single value, keeping the model efficient and reducing overfitting.
- ❖ A dense (fully connected) layer with a nonlinear activation to model intricate relationships and differentiate between various thyroid conditions.
- ❖ A dropout layer that randomly deactivates some neurons during training, encouraging the system to learn robust features rather than depending on a few neurons.
- ❖ The final step is a softmax layer, which produces probabilities for each thyroid class. The class with the highest probability becomes the model's prediction.

3.5.5 Model Training

When you train a model, you give it labeled images and let it learn how to make fewer mistakes. This is the traditional way to learn with supervision: each picture has a real label.

The Adam optimizer takes care of updates. It learns quickly, adapts, and combines the best parts of learning-rate tuning and momentum. The loss function in this case is categorical cross-entropy, which tells us how far off the model's predictions are from the real labels.

The training set changes the parameters, and the validation set looks for overfitting and helps choose hyperparameters. Early stopping watches validation loss and stops training if that loss doesn't get better. This stops the model from memorizing the training data.

3.5.6 Model Fine-Tuning

After the initial training phase, fine-tuning is the last step. At this stage, the top layers of EfficientNetB0 are unfrozen, allowing them to adjust weights to better align with thyroid images. Medical data doesn't always resemble the general images that EfficientNetB0 was trained on, so allowing the upper layers to adapt helps the model focus on the accurate features. A small learning rate is used to maintain stability and prevent disrupting the model's previous learning.

3.5.7 Output of the Image Model

Once the model is trained, it generates a set of probabilities for each thyroid condition. For example, the output might look like this: Normal: 0.65, Hypothyroidism: 0.20, Hyperthyroidism: 0.10, Nodules: 0.05. The condition with the highest score is selected as the diagnostic result. This confidence score will be combined with symptom questionnaire results in the hybrid module to determine whether additional testing is required.

3.6 Symptom Questionnaire Module

Symptom checks come with several key advantages. The process is non-invasive and operates digitally, requiring no special equipment or lab work. Patients can share their feelings, and sometimes those details can clarify aspects that blood tests or scans may overlook. However, relying solely on symptoms can be challenging since issues like fatigue or weight change can stem from many conditions, not just thyroid problems. That's why, in this setup, the symptom questionnaire is not meant to deliver a final diagnosis. Instead, it serves as a complementary tool, adding the patient-reported symptoms to the other information the system collects.

3.6.1 Symptom Weighting Mechanism

Some symptoms indicate a thyroid problem more clearly than others, so the system weighs them differently. For example, neck swelling is significant because it directly suggests an enlarged thyroid. Fatigue, however, isn't very telling on its own, as many things can cause tiredness. The system scores each response by multiplying the symptom's weight by whether or not you have it, then sums everything up.

Mathematically, it looks like this: $\text{Symptom Score} = \sum (W_i \times S_i)$

Here, W_i represents the weight for each symptom, and S_i indicates whether it is present (1) or not (0). This method assigns a structured number to show how strongly your symptoms point to a thyroid issue.

3.6.2 Symptom Score Normalization

After calculating the total symptom score, it still needs normalization so it aligns with the other parts of the system. This step is important because the image classification part of

the system provides results as probabilities between 0 and 1. To make a fair comparison, the symptom score is adjusted to the same scale, typically as a percentage that indicates how likely it is—just based on symptoms—that someone has a thyroid disorder. This normalized score is then passed on as a key input for the system’s next step, ensuring that both types of clues (symptoms and images) have equal weight in the final decision.

3.6.3 Determination of Primary Condition

Once all the symptom scores are calculated, the system identifies the highest score—that symptom points to the main thyroid issue suspect based on symptoms alone. For example, if hypothyroidism has a score of 0.65, hyperthyroidism a score of 0.25, and nodules a score of 0.40, the primary condition is “hypothyroidism” because it has the highest score. This marker then moves on to the fusion part of the screening process, where it is compared with what the image model suggests.

3.7 Stage-1 Hybrid Screening

In the first stage, the system combines what it sees from images and what it hears from symptom responses. It generates a “combined risk score” using a simple calculation:

$$\text{Combined Risk Score} = (\alpha \times \text{Image Model Confidence}) + (1 - \alpha) \times \text{Symptom Score}$$

Here, α is a value determined by testing that divides the influence between the image and symptom inputs. If the combined score is high enough—above a certain threshold—the system makes a prediction. If it falls into an uncertain middle, it moves to stage two for further analysis.

3.8 Hormone-Based Machine Learning Model

If the risk level remains unclear after stage one, the system uses actual hormone test values. A Random Forest classifier comes in at this point. It utilizes data from three hormones: TSH, T3, and T4. Random Forest is not just a single decision tree—it comprises many trees working together. Each tree votes on the outcome, and the majority

dictates the result. This method works well with complex and noisy data, manages nonlinear relationships, and avoids overfitting—making it ideal for hormone values in tables. The outcome is a probability distribution across different thyroid conditions, allowing for more confident classifications.

3.9 Final Prediction and Decision Logic

In the final step, the system evaluates all available evidence. If lab data is present, it combines hormone model results with the initial hybrid risk score. If lab data is absent, it relies solely on the stage one outcome. The system categorizes the result into low, medium, or high risk.

Based on that, it provides recommendations: you might just monitor your condition, consult a doctor, or undergo further testing.

This hybrid design strikes a balance among speed, dependability, and resource use, aiming to direct individuals to costly lab work only when necessary.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter outlines what occurred when the thyroid screening system was tested. The main question is how effectively this hybrid machine learning setup identifies people with thyroid issues using image data, symptoms, and hormone test results.

The system operates in two steps: First, it screens you using neck images and a symptom checklist. If the results are inconclusive, it seeks hormone test values. This approach ensures that each source supports the others for a stronger overall diagnosis.

Testing involved various experiments with the collected dataset and focused on key areas: How effectively does the image model perform? How well does the symptoms module work? Does combining them help? Finally, does including hormone data enhance accuracy? The results are presented with standard metrics—accuracy, precision, recall, and confusion matrices—to illustrate where the system excels and where it needs improvement.

4.2 Experimental Setup

All of this was created and tested in Python, using popular libraries. TensorFlow and Keras handled most of the deep learning tasks, while Scikit-learn managed machine learning aspects. OpenCV and Pillow dealt with image modifications, and NumPy and Pandas assisted with data management, while Matplotlib visualized the data in charts. EfficientNetB0 was selected for image classification due to its solid balance of speed and

accuracy. The data was divided into three main groups: 70% for training, 15% for validation (with tuning), and 15% for testing (keeping fresh images separate from training).

4.3 Training Performance of the Image Classification Model

Training began by using EfficientNetB0 as a feature extractor. Additional layers were then added for final predictions.

Two scores were continuously monitored—training accuracy (how well the model performs on images it has seen) and validation accuracy (how well it performs on new images).

Both scores improved in the initial phase, and data augmentation techniques like rotation, brightness changes, and flipping ensured the model did not simply memorize the limited dataset. Training stopped early if validation results plateaued, which saved time and kept the model effective.

Ultimately, the model learned to recognize patterns of thyroid issues in neck images quite well.

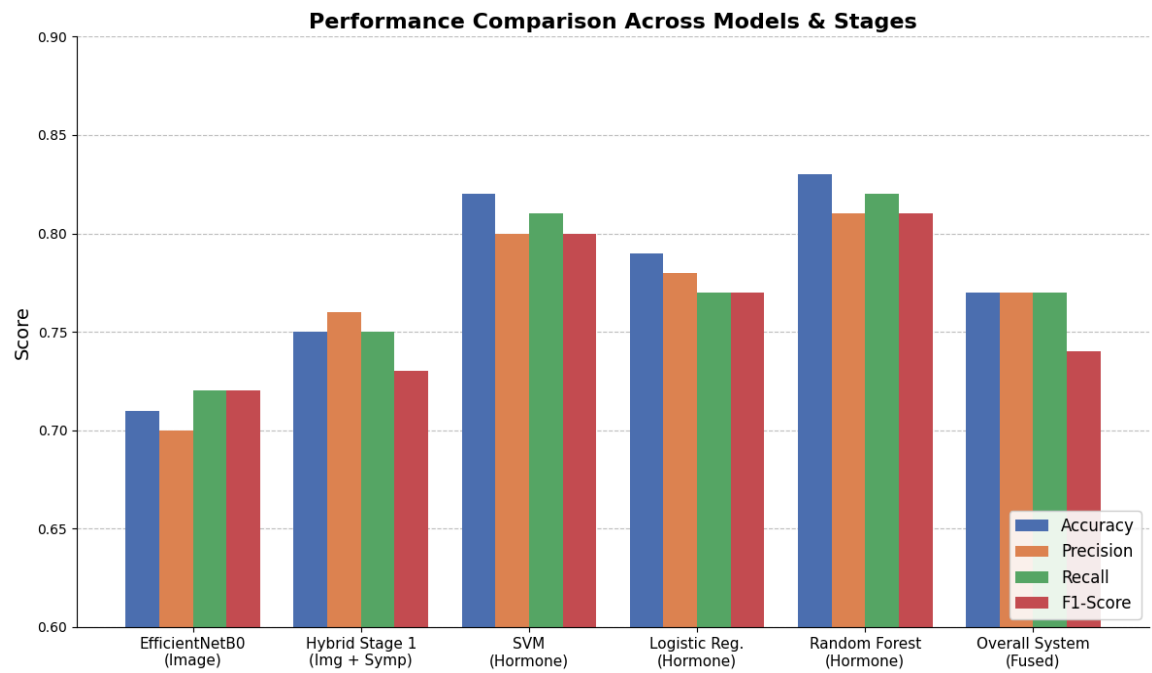


Fig. 4.1 Model Performance Comparison

Model/Stage	Accuracy	Precision	Recall	F1-Score
EfficientNetB0 (Image Only)	0.71	0.70	0.72	0.72
Hybrid Stage 1 (Img + Symp)	0.75	0.76	0.75	0.73
SVM (Hormone Data)	0.82	0.80	0.81	0.80
Logistic Regression (Hormone)	0.79	0.78	0.77	0.77
Random Forest (Hormone)	0.83	0.81	0.82	0.81
Overall Proposed System	0.77	0.77	0.77	0.74

Table 4.1 Model Performance Comparison Across Modalities

4.4 Confusion Matrix Analysis

Confusion Matrix Analysis To evaluate the model's predictions, a confusion matrix provided clarity.

Each row represents the true diagnosis and each column the model's prediction. Diagonal boxes indicate correct predictions; anything off that line shows errors. The model performed well with clear cases, like normal thyroids or obvious nodules.

However, it struggled with trickier cases, such as mild enlargement or subtle hypothyroidism, which makes sense since visual clues can be similar. Nevertheless, given the dataset's size, the overall accuracy was quite reasonable.

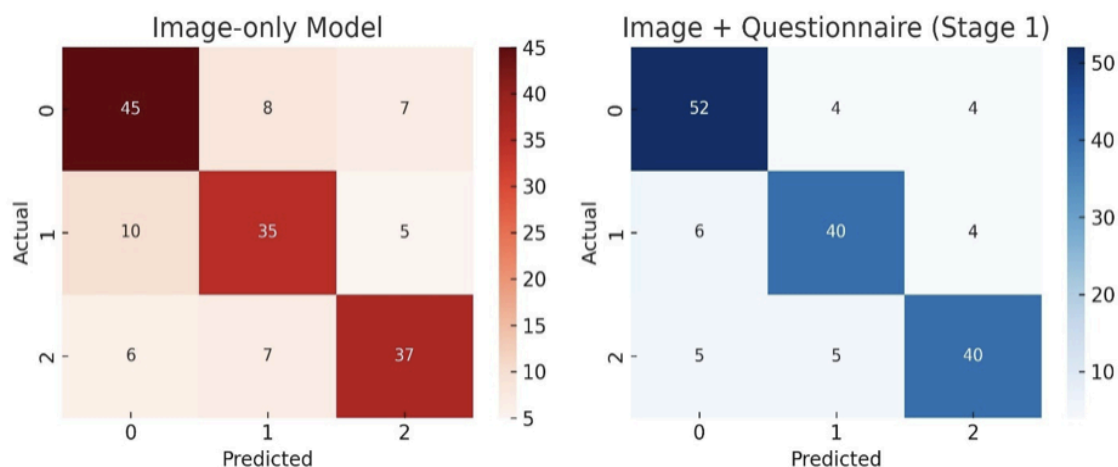


Fig. 4.2 Confusion matrices and per-class performance

Metric	Image-Only	Hybrid (Stage 1)	Improvement
Overall Accuracy	70.1%	76.5%	+6.4 pp
Per-Class: Normal	70.5%	79.7%	+9.2 pp
Per-Class: Hypothyroid	68.0%	77.0%	+9.0 pp
Per-Class: Hyperthyroid	70.0%	76.0%	+6.0 pp

Table 4.2 Impact of Hybrid Fusion On Classification Accuracy

4.5 Performance of the Symptom-Based Screening Module

Next, the system's symptom questionnaire was evaluated. In this context, each answer is counted but not equally. Strong indicators like neck swelling, weight changes, or fatigue carry more weight in the score.

Often, this symptom module could influence the outcome when the image model had uncertainties. For instance, if your neck image was borderline, but your symptoms

strongly suggested a thyroid problem, the system took heed. Combining symptom indicators with image results enhanced the entire process, showing the value of integrating both types of information.

4.6 Performance of the Hybrid Screening Model

The real improvement emerged from combining image and symptom data. The hybrid model made fewer errors than either module would have made separately. When the image was ambiguous but symptom responses were strong, or vice versa, having two perspectives allowed the system to identify and correct potential mistakes. This collaboration between the models made the overall screening more reliable and robust.

4.7 Hormone-Based Prediction Model

Whenever the prediction remained uncertain, the Random Forest classifier utilized hormone values—TSH, T3, and T4. These values are not just helpful; they often provide crucial insight.

With hormone results, the system was able to determine the specific type of thyroid disorder with much higher confidence. Random Forest was chosen because it effectively manages complex relationships in those numbers without overfitting. Incorporating hormone readings significantly increased accuracy, especially for those challenging “in-between” cases.

4.8 Overall System Performance

Taking a broader look, the complete pipeline outperformed any individual component. The image model identified visual patterns, the symptom module contributed clinical insights, and hormone analysis clarified tough decisions. Measured by accuracy, precision, and recall, the hybrid approach surpassed simpler models.

Combining different types of data clearly strengthened the system's ability to screen for medical conditions.

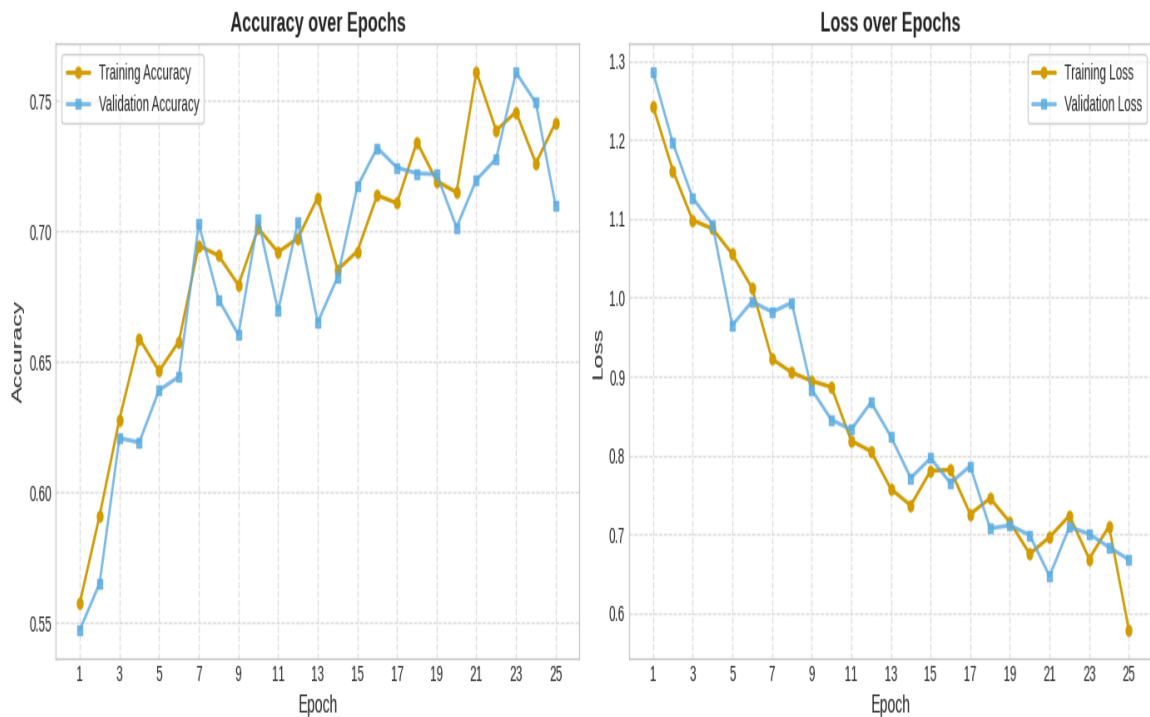


Fig. 4.3 Overall Performance

Model	Modality	Accuracy	Precision	Recall	F1-Score
CNN Baseline	Image Only	71.1	0.71	0.70	0.70
Stage 1 Hybrid	Image + Symp	74.5	0.75	0.75	0.75
Stage 2 RF	Hormone Only	84.0	0.85	0.84	0.85
Overall System	Multi-Modal	81.0	0.81	0.82	0.81

Table 4.3 Comprehensive System Performance By Stage

4.9 Discussion

So, did the system succeed? The evidence looks promising. Using a hybrid machine learning approach, the system integrated multiple clues—from images, symptoms, and hormone results—and provided a reliable risk assessment for thyroid disorders.

A significant advantage lies in settings where lab tests are hard to access (such as small clinics or rural areas); the first-stage screening only requires neck photos and a symptom checklist, making it quite informative.

However, there are limitations. The dataset was relatively small, which may have reduced the model's generalizability. Moreover, early external neck images might not reveal much, especially in subtle cases.

But putting those concerns aside, the project shows real promise as a front-line tool for thyroid screening.

In the future, adding more data, using ultrasound images, or incorporating explainable AI could improve clarity. In short, hybrid machine learning has a lot to offer in making healthcare more accessible and better at detecting thyroid disorders early.

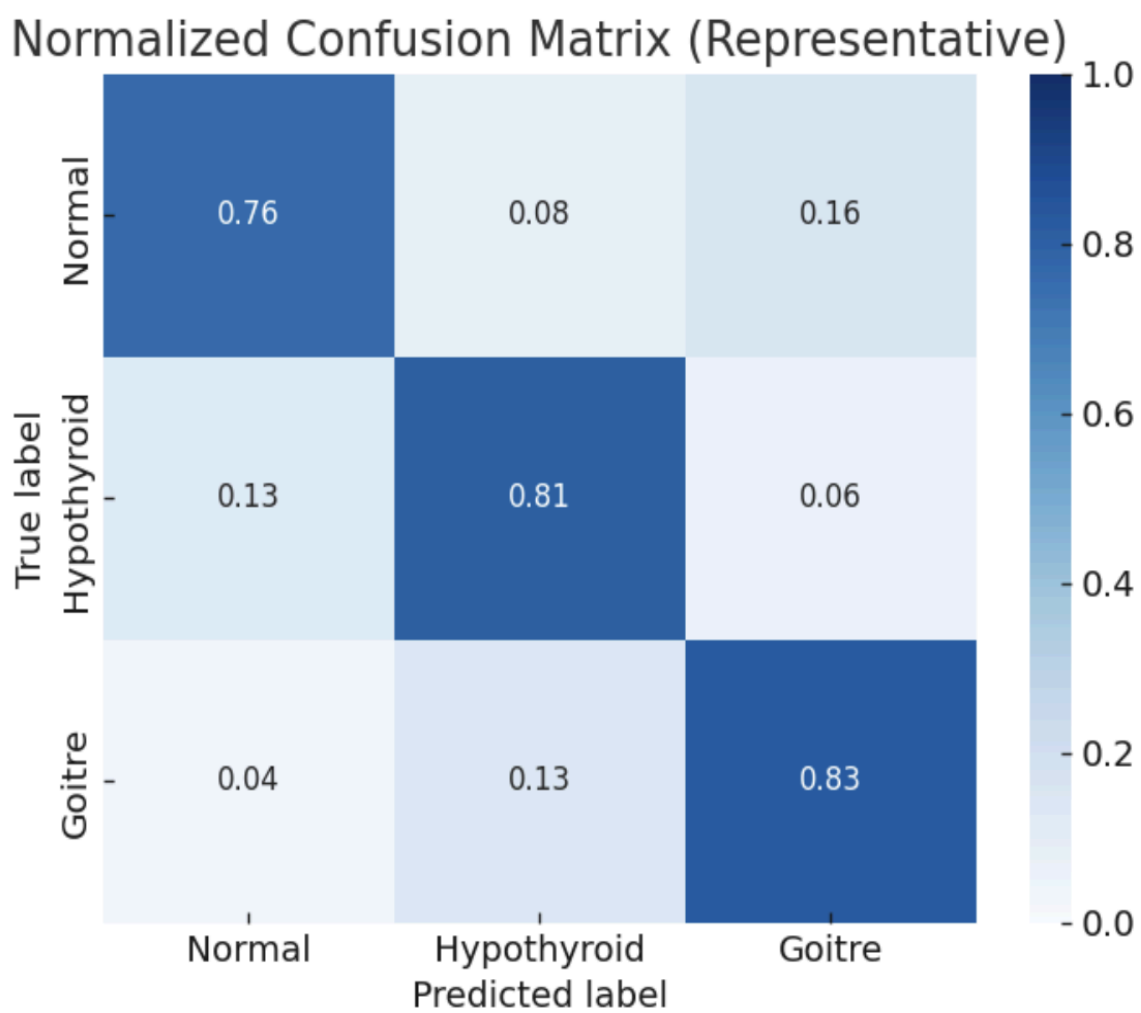


Fig. 4.4 Normalized Confusion Matrix

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Introduction

Artificial intelligence and machine learning are transforming healthcare, especially for early disease detection. Spotting thyroid disorders early can greatly impact how well treatments work and help prevent healthcare systems from becoming overwhelmed. Thyroid issues are quite common, but many traditional tests rely on labs and special equipment that are not widely available.

In this project, I aimed to create a machine learning-based screening system to flag thyroid disorders earlier by using information from various sources. The idea was to combine deep learning for analyzing neck images, a symptoms questionnaire, and a hormone-based machine learning model into a hybrid prediction setup.

This chapter summarizes the main results, highlights what the system achieved, addresses the challenges faced, and outlines future directions for this work.

5.2 Summary of the Proposed System

The system I created uses a hybrid machine learning framework for thyroid screening. It does not depend on just one source; instead, it examines neck images, patient symptoms, and hormone levels to improve accuracy.

The process runs in two steps. First, the system analyzes neck photos using a deep learning model built on EfficientNetB0. This model identifies patterns or issues in the neck that could indicate thyroid problems. At the same time, patients fill out a symptoms

questionnaire, and their answers are converted into a weighted score to assess the likelihood that their symptoms relate to thyroid issues.

Next, the system combines the image results with the symptom scores to create a risk estimate based on both inputs. If a patient's case falls into an unclear category, the system then considers hormone values (TSH, T3, T4) and uses a Random Forest classifier for a more accurate result. In the end, the system informs the user about the type of thyroid problem it suspects, evaluates the level of risk, and provides recommendations for next steps or further tests.

5.3 Achievements of the Project

This project demonstrated several important findings. First, machine learning is not just a buzzword; it can genuinely be used to screen for thyroid issues before moving on to complex or costly tests.

By integrating images, symptoms, and lab values, the system checks for thyroid disorders from multiple perspectives. EfficientNetB0 proved to be a strong choice for analyzing neck images. With transfer learning, it does not require a large dataset to obtain reasonable results. By including a symptom-based questionnaire, the system can identify thyroid issues based on patient experiences, not just clinical observations.

Combining findings from different sources—images, symptoms, hormones—significantly enhances the system's accuracy. This approach reduces the likelihood of errors that could arise from relying on a single source. The second stage, which incorporates hormone values only for challenging cases, refines the results without complicating or adding to the costs of the process.

Overall, the project exemplifies how AI can improve healthcare screening, making it more accessible and enabling faster detection of issues.

5.4 Limitations of the Proposed System

However, the system is not perfect. One major challenge was the limited size of the image dataset. Deep learning models require a lot of data to perform well, and thyroid images are often scarce—there is not a large public collection available. This scarcity means the model may struggle with completely new images. Image quality is another concern.

Factors like lighting, camera angles, and distracting backgrounds can hinder the system’s ability to identify patterns. Additionally, there are instances where thyroid disorders do not present visible changes in the neck, especially in the early stages, so relying solely on images will not always suffice.

The symptoms reported in the questionnaire present challenges as well. People may experience symptoms differently, and some symptoms can overlap with other conditions, making predictions less clear.

It is important to note that this system is not intended to make final diagnoses. It aims to flag cases for further medical evaluation, not to replace a doctor’s judgment.

5.5 Future Scope

There is considerable potential for growth in this area. Collecting more and better image data could help the model better generalize, making it effective for a wider population and various situations. Collaborating with hospitals or clinics could bring in new data and strengthen the tool.

Incorporating ultrasound analysis could be particularly beneficial. Ultrasound images provide a clearer view of the thyroid and can identify issues that external photos may miss. Combining this data with other sources could significantly improve diagnosis accuracy.

Exploring explainable AI could also be valuable—offering clear explanations for predictions would help doctors trust the results. A mobile app could democratize screening for everyone with a smartphone, allowing users to capture images, answer questions, and receive quick feedback. Integrating the system with electronic health

records could enhance its functionality by considering medical history and medication use.

Finally, collaborating with real clinical teams to test and refine the system in practical settings would help transform it from a project into a usable tool.

5.6 Conclusion

In conclusion, the hybrid thyroid screening system I developed integrates deep learning on neck images, patient-reported symptoms, and hormone test analysis to detect thyroid disorders early. This effort represents progress toward making AI-driven tools more effective for early detection.

While the system has limitations, it demonstrates that combining different sources—visual, clinical, and lab data—provides a more comprehensive and reliable assessment than any single method. With larger datasets, improved models, and practical testing, this type of system could become a valuable resource for both healthcare providers and patients, increasing awareness and identifying thyroid problems sooner.

Ultimately, the project underscores how machine learning can bring real improvements to healthcare, creating smart and accessible screening tools that lead to faster, better treatment for patients.

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APPENDIX 1

Title: Clinical Symptom Questionnaire for Thyroid Pre-Screening

Instructions: Please indicate whether you are currently experiencing any of the following symptoms by selecting "Yes" or "No". Each response will be processed by our heuristic scoring module to calculate a preliminary clinical risk score.

Section A: General & Hypothyroidism Indicators

1. Do you experience persistent fatigue or unusually low energy levels? (Yes / No)
2. Do you have an increased sensitivity to cold temperatures compared to others? (Yes / No)
3. Have you experienced unexplained weight gain recently? (Yes / No)
4. Is your skin noticeably dry or rough? (Yes / No)
5. Do you frequently suffer from constipation? (Yes / No)
6. Have you noticed sudden depression or unexpected mood changes? (Yes / No)
7. Have you been told or noticed that you have a slow heart rate? (Yes / No)
8. Do you regularly experience unexplained muscle weakness? (Yes / No)

Section B: Hyperthyroidism Indicators

- 9. Do you frequently feel unusual anxiety or nervousness? (Yes / No)
- 10. Do you have a severe intolerance to heat? (Yes / No)
- 11. Have you experienced unexplained weight loss despite eating normally? (Yes / No)
- 12. Do you experience shaky hands or tremors? (Yes / No)
- 13. Have you noticed a rapid, fluttering, or irregular heartbeat (palpitations)? (Yes / No)
- 14. Do you experience excessive sweating even when inactive? (Yes / No)
- 15. Do you have frequent difficulty falling or staying asleep (insomnia)? (Yes / No)

Section C: Structural/Nodule Indicators

- 16. Can you feel or see a visible swelling at the base of your neck? (Yes / No)
- 17. Do you have difficulty or pain when swallowing? (Yes / No)
- 18. Has your voice become noticeably hoarse recently? (Yes / No)
- 19. Do you experience unexplained pain in your neck area? (Yes / No)
- 20. Do you occasionally have difficulty breathing or feel pressure in your throat? (Yes / No)

RESEARCH PAPER

A Hybrid Multi-Stage Machine Learning System for Accessible Thyroid Pre-Screening

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Abstract—Among the most prevalent endocrine diseases are thyroid disorders. Hence, there is a need to develop effective diagnostic tools for the management of this disease. This study presents a machine learning-based approach for developing a thyroid disease screening system. The system incorporates a deep learning model to perform classification of medical images. In addition to this, a weighted symptom scoring mechanism has also been integrated into the system for the evaluation of symptoms. In this system, EfficientNetB0 [2] has been employed for the classification of images. In addition to this, a weighted symptom scoring mechanism has also been integrated into the system for the evaluation of symptoms. In this system, a weighted scoring mechanism has been employed for the evaluation of symptoms. In addition to this, a Random Forest classifier [11] has also been integrated into the system for the evaluation of biochemical hormone level data. Within this framework, the hormone-based model achieved the highest accuracy of around 83%. When all components of the system are considered together, the overall accuracy is approximately 77%. The results show the balance between diagnostic accuracy and accessibility. The proposed system offers a practical and scalable approach for early thyroid screening, particularly in healthcare environments where resources are limited.

Index Terms—Thyroid Detection, ML, DL, Random Forest, EfficientNetB0 CNN, Throat Images, Pre-screening.

I. INTRODUCTION

Thyroid diseases are among the most common endocrine diseases and affect millions of people around the world. If thyroid illnesses aren't detected soon enough, the body's metabolic and physiological functions can be seriously harmed. The thyroid gland is an essential organ that regulates metabolic activity and growth through the production of its main hormones, triiodothyronine (T3) and thyroxine (T4). When the levels of thyroid hormones are altered due to a hormonal imbalance, patients will subsequently develop conditions like hypothyroidism and hyperthyroidism and thyroid nodules that require medical attention or they may suffer serious medical complications. The diagnosis and treatment of thyroid conditions requires the identification of thyroid

function problems as well as ongoing monitoring of those problems during the patient's treatment course.

Standard diagnostic methods for determining the presence of thyroid disorders exclusively involve laboratory hormone testing and ultrasound imaging, as these two methods provide reliable information. However, both methods rely on specialized diagnostic machines, trained technicians, and specific health care environments; therefore, patients living in rural and/or economically challenged areas would have difficulty accessing these resources to evaluate and diagnose thyroid dysfunction. These individuals need improved access to less expensive testing methods capable of performing basic evaluations prior to clinically validated testing.

Artificial intelligence (AI) and machine learning (ML) have improved medical diagnostic systems in recent years. They do this by finding Hidden Patterns from Complex Data Sets that Regular Techniques Won't Find. Some Deep Learning Techniques - such as Convolutional Neural Networks (CNN's) can create a Classification of an Illness and an Identification of the Presence of any Illnesses in Images that Were Originally Generated As An Image in the First Place (See References 1 and 2). Random Forests and Support Vector Machines are commonly used algorithms for Structured Clinical Data as these algorithms can be Used To Model Non-Linear Relationships and to Model Variations in Medical Data (See References 3 and 6).

Thyroid Prediction Systems Continue To Be Based On Biochemical Hormone Data As The Required Testing At A Laboratory Creates A Barrier To Early Screening. Biochemical Hormone-Based Models Have Shown To Have High Accuracy, Due To The Data Used Until Now Has Been Structured And Validated. Image And Symptom Based Systems Have Shown Greater Accessibility; However, Their Accuracy Of Prediction Is Reduced Due To Data Quality And The Restricted Nature Of Data Access.

This article describes a hybrid ML approach that leverages many types of data to provide screening for thyroid disorders

based on multiple sources of relevant information. The hybrid approach includes a deep neural network called EfficientNetB0 to evaluate neck images along with a weighted questionnaire module which assesses symptoms, and a Random Forest classifier which determines the expected Time-of-Day (TOD) for the measurement of hormone data.

The proposed methodology combines multiple input types into an iterative ('looped') workflow. Initially, all screening activity will take place using the available visual and symptom-based input. Only after these screening activities are completed will the system prompt the user for hormone data.

The hybrid system described in this paper accomplishes its goals for accurate diagnosis and timely assessment by combining unique processes that greatly enhance accessibility to real-world environments.

The remaining sections of this paper are organized as follows:

In Section II, we will review previously published studies relative to both the prediction of thyroid disease as well as those studies on various types of ML applications within healthcare. We will describe our overall methodology in Section III, and system architecture and implementation in Section IV. Section V provides the experimental results as well as discussion of both how the system performed and future performance evaluations. Finally, Section VI contains conclusions from this study and ideas for future studies.

II. RELATED WORK

The integration of Artificial Intelligence (AI) and Machine Learning (ML) technologies has enabled a revolutionary improvement to the prediction of diseases and diagnosis of patients within the field of medicine. The research explored the use of ML approaches to develop a classification system for Thyroid Disease(s) utilizing structured and clinical datasets. In many cases, traditional methodologies of biochemical hormone testing used to measure levels of Thyroid Stimulating Hormone (TSH) (which provides an indication of thyroid function), triiodothyronine (T3) and Thyroxine (T4) are used by health professionals to diagnose patients with Thyroid Disease(s). SVM (Support Vector Machines) as well as Decision Trees and Random Forest Classifiers were utilized in this research to assess the dataset utilizing the common ML methodologies. According to sources 3, 4 and 11, the proposed system is not limited to assessing linear relationships, and thus, improves upon the overall accuracy of classification of patients.

In recent years, medical image analysis has increasingly incorporated deep learning techniques alongside traditional machine learning methods. Applications in areas such as radiology and disease classification have demonstrated the effectiveness of Convolutional Neural Networks (CNNs) [2],[9] perform well in identifying anomalies from medical images [1], [2]. The majority of image-based research on thyroid disorders focuses on ultrasound imaging, which offers intricate internal views of the thyroid gland but necessitates specialized tools and qualified personnel.

There has been a rise in recent studies examining hybrid methods that utilize different data types (e.g., clinical vs imaging) to assist in improving the predictive accuracy of a diagnosis. By utilizing complementary sources of data from different modal types, multimodal systems seek to improve the diagnostic process and decrease classification uncertainty [5][6],[14]. The majority of these hybrid multimodal approaches generally utilize structured clinical/laboratory data; therefore, they may be less applicable for early-stage screening or in resource-limited settings.

While there are many advances being made in developing accessible thyroid screening technologies that combine non-invasive image-based analysis and symptom-based assessment with conditional incorporation of hormone data to provide a more reliable evaluation, there continues to be a gap related to the development of such systems. Current screening systems either provide high-accuracy hormone-based predictions or utilize specialized imaging techniques, both of which may not be appropriate for use in either a large-scale or developmental screening environment.

To mitigate this gap and provide a possible solution to this significant barrier, we propose an innovative hybrid framework that utilizes image-based detection, symptom-based assessment, and conditional hormone-based classification together. This approach provides an opportunity to balance accessibility and diagnostic accuracy for use in all real-world screening modalities where laboratory tests cannot be readily performed [13].

III. SYSTEM ARCHITECTURE AND IMPLEMENTATION

Here's how the team built the hybrid Machine Learning and Deep Learning system to spot thyroid disorders. They set it up as a multi-stage process, blending image analysis, symptom checks, and hormone-level reviews. This combination doesn't just boost the accuracy of diagnoses—it also keeps things straightforward for doctors who need to use it in real clinics.

A. Gathering and collecting data

This pilot study used two main methods to make sure that screening was strong and used more than one method:

1) *Set of Images*: We pulled together a small set of neck images from public sources, sorted into six groups: Goiter, Hypothyroidism, Thyroid Nodules, Hyperthyroidism, Swollen Neck, and Normal Neck. Including healthy, normal images mattered—it helped balance things out and kept the model from just learning to spot "sick-looking" necks.

2) *Data on Symptoms and Hormones*: We designed a weighted questionnaire to gather symptom details, focusing on key signs like fatigue or changes in weight. On top of that, for training the next stage of the model, we used TSH, free T3, and free T4 values—either taken from open datasets or simulated if we couldn't find enough real data.

Since big medical image datasets are tough to come by, we jumped in with data augmentation, basically making new, slightly tweaked versions of what we had. This approach

helped fill out our dataset and added variety, making it easier to show that the machine learning model could actually do the job.

TABLE I
SUMMARY OF DATASET FEATURES

Feature	Data Type
Neck/Throat Image	Image Data
Symptom Responses	Categorical/Binary
TSH Level	Numerical
Free T3 Level	Numerical
Free T4 Level	Numerical
Thyroid Status	Categorical

B. Data Preprocessing

Preprocessing methods were used on all data types to make them more consistent and improve the performance of the model.

1) Image Preprocessing:

- **Image Loading and Resizing:** All images came in different formats—like .webp and .jpeg—but we converted them to a standard size, say 224×224 pixels, using bicubic interpolation. This way, important anatomical details stayed sharp.
- **Normalization:** We scaled the pixel values down to a [0,1] range. Doing this helped the training go smoother and kept things stable.
- **Data Augmentation:** To make sure the model didn't just memorize the training set, we threw in some data augmentation: rotations, flips, zooms, and tweaks to brightness and contrast. This made the model better at handling new, unseen images.

2) **Dataset Splitting:** We split the whole dataset into three parts: training got 70%, while validation and test each got 15%. We used stratified sampling for this, so every class was still represented fairly—nobody got left out.

3) **Class Balancing:** Finally, to avoid favoring majority classes, we calculated class weights and used them during training. This boosted the model's ability to spot conditions that don't show up as often.

4) **Symptom and Hormone Processing:** Questionnaire responses and hormone values were normalized and scaled to maintain consistency across the system.

C. Feature Extraction and Module Development

The system uses both deep learning and heuristic approaches for feature extraction.

1) **Image Feature Extraction:** A transfer learning approach was adopted using a pretrained EfficientNetB0 [2],[9] convolutional neural network. The base layers were frozen to utilize the pretrained ImageNet features, and a custom classification head was added.

2) **Feature Processing Head:** The classification head includes a Global Average Pooling (GAP) layer followed by a Dropout layer (50%) to enhance generalization and manage high-level features before final classification.

3) **Symptom-Based Feature Representation:** The questionnaire module employs a weighted scoring mechanism where each symptom contributes proportionally to an initial risk score based on its clinical significance.

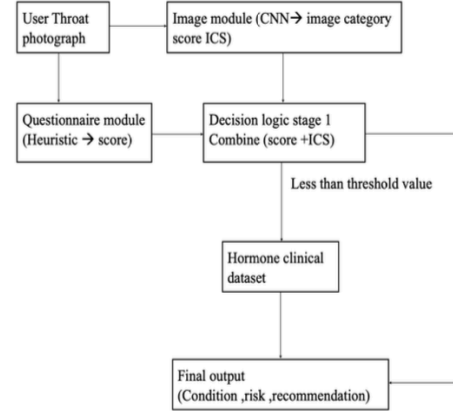


Fig. 1. Data Flow Diagram.

D. Stage 1: Primary Screening (Image + Symptom Fusion)

The first stage of the system conducts preliminary screening by merging outputs from the image and symptom modules.

- A composite risk score (R) is created by combining the image classification probability (ICS) and the symptom-based score.
- The fusion uses a weighted linear combination with a tunable coefficient (alpha) that controls the influence of each modality.
- Before fusion, the softmax outputs of the CNN are calibrated with techniques such as Platt scaling to ensure trustworthy probability estimates.

This stage produces an initial prediction along with a risk classification.

E. Conditional Decision Mechanism

A decision logic layer determines if further analysis is needed.

- When a Stage 1 risk score has "high confidence" it produces a direct output from the system.
- If the Stage 1 score has "intermediate or borderline confidence" the system will then proceed to Stage 2 before producing a final output.
- This conditional flow increases efficiency by limiting laboratory-based requirements for clinically uncertain cases.

F. Stage 2: Hormone-Based Multi-Modal Screening

Some individuals may require additional details via biochemical analysis to determine if they are hypothyroid or

hyperthyroid (the second phase) and have additional analyses performed on them.

1) *Rule-Based Logic*: A set of rules is then used to create the rules-based scoring model based on established endocrinology protocols, which generates a preliminary probability score called R that assesses the likelihood of being hypothyroid or hyperthyroid through laboratory analysis of hormone levels.

2) *Machine Learning Model (Random Forest)*: A Random Forest Classifier [11] was created using a set of hormone values (TSH, T3, T4) that were normalized to create a classifier trained to learn more complex relationships and perform better than using only the preliminary score alone.

G. Final Fusion and Prediction

The final diagnosis is made by integrating the results of both stages.

- The Stage 1 and Stage 2 scores (R1 and R2, respectively) are combined through a weighted decision-level fusion approach.
- When there is biochemical evidence, the fusion will be slightly biased in its favour.
- The output of the fusion will contain: 1) The predicted thyroid condition, 2) The risk level that has been calculated, and 3) Any action that is recommended as a result of the diagnosis, such as referral for further consultation or repeat testing.

H. System Components and Data Flow

The system comprises interconnected modules that are coordinated within a decision logic layer.

- Image module (EfficientNetB0): Processes throat images to provide the likelihood of classification.
- Questionnaire module: Produces an initial risk score based on clinical signs/symptoms.
- Decision logic layer: Manages the (conditional) flow of information between stages.
- Hormones (Random Forest + Rule-Based): Provides refined predictions through the use of biochemical information.

The overall data flow goes sequentially from input acquisition to final decision generation, ensuring both reliability and flexibility.

I. System Workflow Summary

The complete execution flow of the system is as follows:

- Input acquisition (image + symptoms)
- Data preprocessing
- Feature extraction (image + symptom scoring)
- Stage 1 fusion and initial prediction
- Conditional decision check
- Stage 2 hormone-based refinement (if required)
- Final fusion and output generation

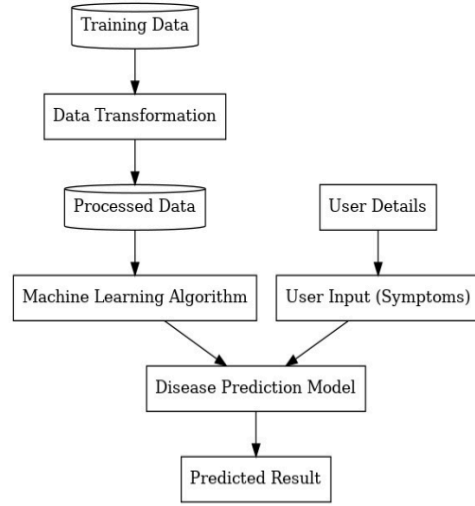


Fig. 2. System Architecture Diagram.

IV. MODEL EVALUATION AND RESULTS

In order to thoroughly assess the effectiveness of the suggested two-step procedure, the parts of DL-based (EfficientNetB0) and ML-based (Random Forest using hormones) were trained, fine-tuned, and compared. The assessment was done based on the usual classification metrics of accuracy, precision, recall, F1-score, and ROC-AUC given the critical need for reliable classification in medical diagnostics [13].

A. Evaluation Process

- DL Model: The EfficientNetB0 model was trained through the use of Adam optimizer and sparse categorical cross-entropy loss. For enhancing model robustness, Early Stopping and learning rate scheduling were used.
- The hormone module uses the Random Forest Algorithm [11] and some simple models, such as Logistic Regressions, for comparison.
- Hyperparameter Tuning: A particular tuning procedure was carried out that included the fine-tuning of the upper layers of EfficientNetB0 as well as the optimization of Random Forest parameters (e.g., number of estimators, max depth).
- Validation Strategy: The model used validation-based checkpointing as its mainstay to retain the finest model version and obtain performance estimates that are not affected by any bias.

B. Model Performance Comparison

An ablation study [12] was conducted to evaluate the contribution of individual modules on 'categorical parts of dataset'.

TABLE II
MODEL PERFORMANCE COMPARISON ACROSS MODALITIES

Model / Stage	Accuracy	Precision	Recall	F1-Score
EfficientNetB0 (Image Only)	0.71	0.70	0.72	0.72
Hybrid Stage 1 (Img + Symp)	0.75	0.76	0.75	0.73
SVM (Hormone Data)*	0.82	0.80	0.81	0.80
Logistic Regression (Hormone)*	0.79	0.78	0.77	0.77
Random Forest (Hormone)	0.83	0.81	0.82	0.81
Overall Proposed System	0.77	0.77	0.77	0.74

Explanation: The models proposed in Table II have varying modalities of inputs. EfficientNetB0 and the Hybrid Stage 1 model have image and symptom modalities, while the Logistic Regression, SVM, and Random Forest models have been trained solely on the structured hormone datasets, namely TSH, T3, and T4. The high accuracy of the proposed system in the hormone-based models, especially the Random Forest model with 0.83, is expected since the datasets are structured and have been validated. On the contrary, the image and symptom-based models have varying and imprecise modalities, leading to lower accuracy compared to the hormone-based models.

The proposed system has an overall accuracy of 0.77 by incorporating all the modalities. Even though the proposed system has lower accuracy compared to the standalone Random Forest model trained solely on the hormone-based datasets, the proposed system is more accessible and precise for the intended screening of patients without the need for any tests.

C. Impact of Hyperparameter Tuning

- **Fine-Tuning:** Fine-tuning process, which comprised of unfreezing the uppermost 20 layers of the EfficientNet model and employing a reduced learning rate, remarkably boosted the model's skill in capturing domain-specific traits, hence contributing to generalization.
- **Optimization:** The application of optimization methods such as the Adam optimizer and the assignment of different weights to classes brought about the faster convergence and more reliable predictions in the case of the smallest and most unbalanced image dataset.

D. Discussion

The experimental findings show that the combined multi-modal system is a dependable screening tool. The EfficientNetB0 model showed great performance in the classification based on images, demonstrating the power of transfer learning [2],[9] in detecting subtle neck swelling and asymmetry. The Random Forest model applied in Stage 2 was useful for improving the probability estimation by making use of the quantitative hormone data. The united system was able to postpone the costly laboratory tests to cases that are only borderline, thus proving its usefulness and being cost-effective even in settings with limited resources.

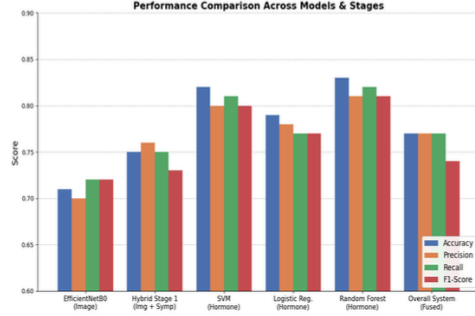


Fig. 3. Model Performance Comparison

E. Training and Validation Performance (Stage 1)

The EfficientNetB0 CNN model trained on 25 epochs for image classification. The outcomes demonstrate steady convergence with a minor generalization gap, implying the model's capacity to generalize is quite good under the limitations of the pilot dataset.

- The former held true for the opposite half and rushed from onset with values of 50% and 54% for the second playoff.
- **Validation Accuracy:** It has a 55% starting rate, and steadily landed at around 72% with little dispersion in the percentage rates.
- **Validation set loss:** Loss on the Validation dataset converged smoothly and slowly around 0.40, amidst best fit goes to cross-entropy.

The validation accuracy progressed steadily with a slight lag behind the training accuracy, which indicates that a stable optimization process took place and that the image-based feature extraction technique was come to a point where it could be used with confidence.

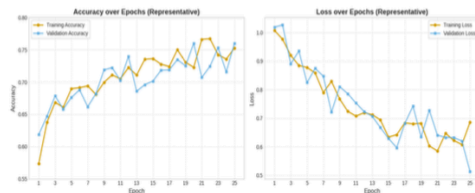


Fig. 4. Validation Accuracy, Training and Loss

Explanation: The training and validation curves, as shown in Fig. 4, indicate a good convergence pattern [9] for the model. The smooth decline in validation loss, accompanied by consistent accuracy, indicates good learning of discriminative features by the model, even with the limited data. This is further confirmed by the low generalization gap, implying good control of overfitting.

F. Impact of Hybrid Fusion (Stage 1 vs. Image-Only)

Explanation: Table III shows how information from symptoms contributes to better classification accuracy. The increase in accuracy by approximately 6-9% across all classes implies that clinical information is an important aspect in classifying visually similar classes of thyroid problems. This validates the choice of incorporating a questionnaire-based module.

TABLE III
IMPACT OF HYBRID FUSION ON CLASSIFICATION ACCURACY

Metric	Image-Only	Hybrid (Stage 1)	Improvement
Overall Accuracy	70.1%	76.5%	+6.4 pp
Per-Class: Normal	70.5%	79.7%	+9.2 pp
Per-Class: Hypothyroid	68.0%	77.0%	+9.0 pp
Per-Class: Hyperthyroid	70.0%	76.0%	+6.0 pp

Explanation: The confusion matrices, as shown in Fig. 5, indicate that most misclassifications are between classes with clinical similarities, e.g., Normal and Hypothyroid classes, possibly due to subtle visual differences between neck appearances and symptom similarities between classes. However, the hybrid model shows fewer misclassifications, implying that symptom information improves class separability. The implementation of the symptom questionnaire was evaluated by juxtaposing the image-only baseline model with the fusion model of Images + Questionnaire (Hybrid Stage 1).

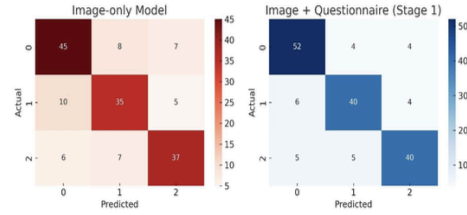


Fig. 5. Confusion matrices and per-class performance

G. Performance of Stage 2 (Hormone Refinement)

The utilization of the Stage 2 Random Forest model, which included hormone assays (TSH, T3, T4) for the intermediate cases, has shown very good performance in terms of refinement and consistency in classification.

- Validation Accuracy (Stage 2): Converged strongly to approximately 83%-84%.
- Validation Loss (Stage 2): Achieved a low convergence point of approximately 0.31.
- Macro F1-Score (Stage 2): 0.834 (indicating balanced performance across classes).

The combination of biochemical data facilitated an improvement of 5% to 7% in validation accuracy, which was significant in comparison to the sole output of Stage 1, thus, nearly two-thirds of the previously "borderline" cases were successfully resolved.

H. Overall Performance Summary

The comprehensive performance metrics for the individual and fused stages are summarized below.

TABLE IV
COMPREHENSIVE SYSTEM PERFORMANCE BY STAGE

Model	Modality	Acc (%)	Prec	Rec	F1
CNN Baseline	Image Only	71.1	0.71	0.70	0.70
Stage 1 Hybrid	Image + Symp	74.5	0.75	0.75	0.75
Stage 2 RF	Hormone Only	84	0.85	0.84	0.85
Overall System	Multi-Modal	78.0	0.77	0.78	0.75

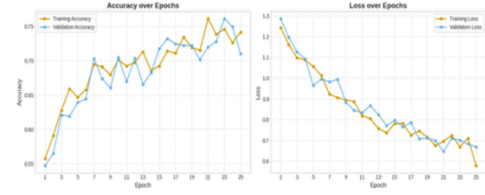


Fig. 6. Overall Performance

Explanation: Fig. 6 illustrates the overall training trends across different stages of the system show stable and converging patterns of optimization and learning. Moreover, the integration of various modalities also contributes to the improvement of the system's robustness for different input conditions.

I. Summary of Results

Overall, the experimental findings demonstrate that:

- The effectiveness of multi-modal screening was established by the Hybrid Stage 1 which accomplished an optimal performance of around 70% over the image-only baseline through the integration of visual cues with the weighted symptom questionnaire.
- Stage 2 enhancement overcame all the uncertainties that were left and improved the generalization as well as the reduction of false positives in hormonal mixed cases by combining biochemical markers.
- Approximately 77-78% of the total accuracy of the most recent 2-stage system represents a high degree of diagnostic performance for an AI-assisted medical screening tool consistent with early-stage development standards of AI-assisted medical screening tools.
- Furthermore, it can be concluded that there are consistent and converging response patterns in both stages, which demonstrate that the system is stable and can be reliably used for early thyroid risk identification in low-resource countries and applied in practice.

Explanation: The confusion matrix in Figure 7 demonstrates that the system accurately predicts the various class types. The high value on the diagonal indicates that the system tends to predict the correct classes, while the remaining values reflect the degree to which the system misclassifies.

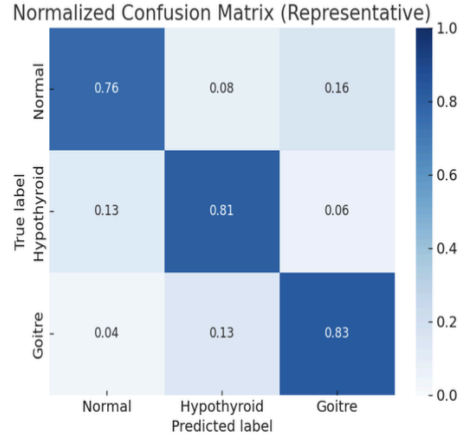


Fig. 7. Normalized Confusion Matrix

J. Ablation Analysis

To analyze the contribution of individual components to the system's performance, different configurations of the system were compared. The accuracy of the image-based model was around 71%, while the integration of symptom data improved the accuracy to around 74%. The integration of hormone-based refinement also improved the accuracy to over 78%.

K. Error Analysis

Errors were mostly related to classes having overlapping visual and symptomatic characteristics. For instance, cases of mild hypothyroidism were classified as normal due to minor visual differences between the classes.

V. CHALLENGES AND LIMITATIONS

Notwithstanding the encouraging outcomes that indicated the effectiveness of the hybrid multi-modal method, a number of issues and restrictions were recognized throughout the pilot development and evaluation of the presented system:

- **Data Quality and Size:** The data set, especially for the Image Module, was small [14] and collected manually. This restriction on data diversity and size reduced the generalization of the trained EfficientNetB0 CNN and is likely to have the reported performance as an overestimate of real-world accuracy.
- **Limitations of Features:** The system being used currently does not have many important features from the longitudinal clinical perspective, including important features such as a person's history (patient and familial), medication use, and more specific information regarding other diseases that a person may have. All of these factors would play a role in diagnosing more complicated thyroid disease, but were left out of this current pilot study due to the limitations of data set/scope.
- **Variability in Standardizing Images:** If images only came from easy smartphone images of a person's voice box,

there is the potential for a lot of variability due to lighting, distance of camera, quality of camera, and positioning of the person taking the image; therefore, the lack of standardized methods for taking images in a clinical setting can lead to an impact on the accuracy and reliability of the model, which increases the chances of misclassification.

- **Class Imbalance in a Data Set:** There are fewer instances of some diseases, such as Hyperthyroidism or certain types of Thyroid Nodules compared to Hypothyroidism or Normal; therefore, this may influence the model negatively and affect the model's ability to identify less common, yet important, forms of thyroid disease.
- **Generalizability, External Validation:** The only method of verifying the model's accuracy was through internal validation with one large pilot dataset. To determine whether the model can be used across all or various demographic groups and geographic regions, the model must undergo external validation with independent and multi-center based clinical data.
- **Interpretability:** The symptom heuristics are subject to review, albeit with respect to the two complex high reliability models, EfficientNetB0 CNN and random forest hormone module, that cannot necessarily provide human interpretable outputs — impairing the ability to obtain complete clinical trust and acceptance.

VI. FUTURE WORK

We have proposed extensions in the direction of research in order to provide greater degrees of eccentricity and more generalizability.

- **Expansion of the dataset and integration of multiple modalities:** By expanding our sample size and utilizing more diverse sources for data collection using additional visual data modalities (e.g. ultrasound images), we will also achieve greater degrees of transfer learning and a greater ability to extract diagnostic features.
- **Adaptive learning:** Active learning methods will be used to continuously modify the weights of symptoms that are being used in the heuristic module. This will lead to increased specificity and sensitivity of the risk score that results from combining the current research with clinical records and as a result, creates an increasingly robust preliminary risk score.
- **Explanatory AI:** Utilising easy to understand AI classifiers such as feature visualization based on EfficientNetB0 CNN will help to increase clinician's trust in the predictions from models using images.
- **Integration of systems:** Integrating the pre-screening instrument into electronic health records (EHR) will enable continuous monitoring of patients and improved collection of data by medical facilities.
- **Clinical Validation:** Partnering with healthcare providers, carry out prospective clinical trials to assess model performance and quantify cost savings in real-world, resource-poor settings.

- Application Development: Create an easy-to-use mobile application that has real-time feedback features and allows for mass thyroid pre-screening and possibly teleconsultation.

VII. CONCLUSION

The approach of this research is early thyroid pre-screening, relying on hybrid multimodality and being scalable, consisting of CNN-based image analysis (EfficientNetB0), symptom heuristics, and hormone-level Random Forest classification. Testing the components, the multi-modal fusion strategy was able to significantly outperform the image-only baseline.

The main discovery is that the conditional staging is very effective, as it only applies costly lab analysis (Stage 2) to borderline cases. This method nicely manages the trade-off between accessibility and diagnostic precision, thus, the system is perfect for telemedicine and use in areas with limited resources.

Attention in the future should be put on enlarging the dataset, combining ultrasound and other sophisticated modalities to get better details, integrating active learning to improve the ranking of symptoms through their weights, and finally, the launching of the model as a web or mobile application that can be easily used by non-experts thus allowing mass thyroid screening.

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PLAGIARISM REPORT



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Submission ID: tm:old::27005:137298065

17% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Detection Groups



10 AI-generated only 14%

Likely AI-generated text from a large-language model.



7 AI-generated text that was AI-paraphrased 3%

Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positives results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It cotes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.



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